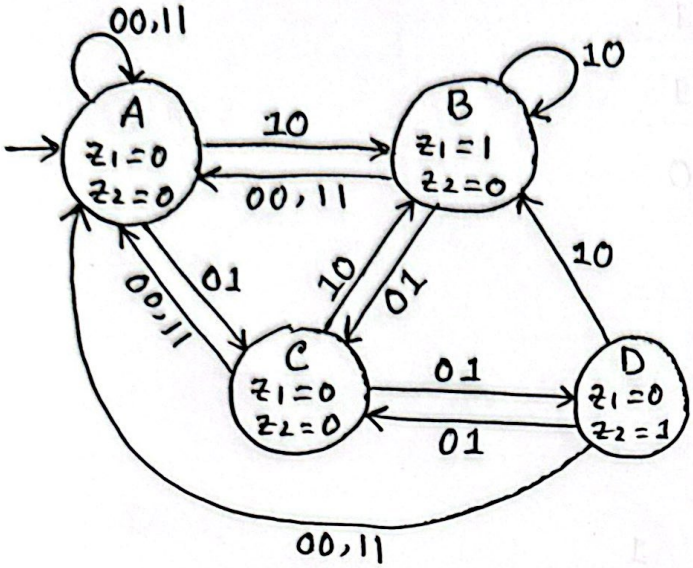
Azmari Sultana

SPRING 25 (A)

① a) Moore type FSM

b)



c)

present state y_2y_1	Next state Y_2Y_1				Output, z z_1z_2
	00	01	10	11	
A (00)	A (00)	C (11)	B (01)	A (00)	00
B (01)	A (00)	C (11)	B (01)	A (00)	10
C (11)	A (00)	D (10)	B (01)	A (00)	00
D (10)	A (00)	C (11)	B (01)	A (00)	01

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d) Number of flip flop = $\log_2(4) = 2$

z_1

$y_2 \backslash y_1$	0	1
0	0	$\boxed{1}$
1	0	0

$z_1 = \bar{y}_2 y_1$

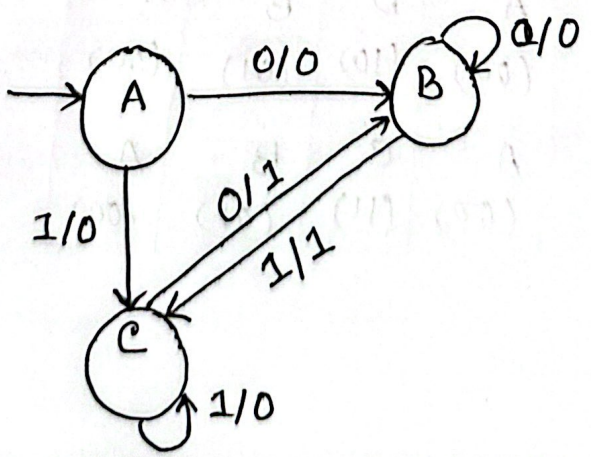
z_2

$y_2 \backslash y_1$	0	1
0	0	0
1	1	0

$z_2 = y_2 \bar{y}_1$

② Mealy - type overlapping → 01 or 10

a)



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b)

Present state y_2y_1	Next state Y_2Y_1		Output, z	
	$w=0$	$w=1$	$w=0$	$w=1$
A (00)	B (01)	C (10)	0	0
B (01)	B (01)	C (10)	0	1
C (10)	B (01)	C (10)	1	0
D (11)	d	d	d	d

c) Y_2

y_2y_1 \ w	0	1
00	0	1
01	0	1
11	d	d
10	0	1

$Y_2 = w$

Y_1

y_2y_1 \ w	0	1
00	1	0
01	1	0
11	d	d
10	1	0

$Y_1 = \overline{w}$

z

y_2y_1 \ w	0	1
00	0	0
01	0	1
11	d	d
10	1	0

$z = y_1 w + y_2 \overline{w}$

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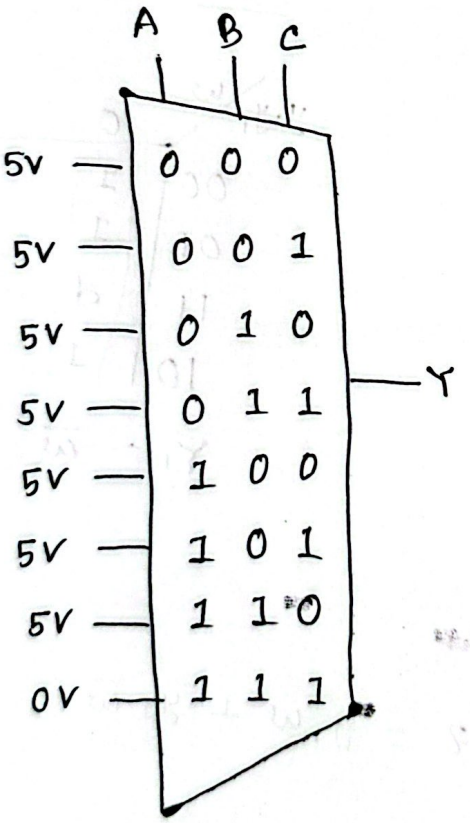
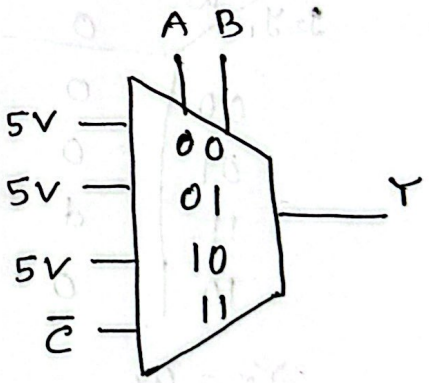
③ a)

A	B	C	Output (Y)
0	0	0	1
0	0	1	1
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	0

8 : 1 MUX



4 : 1 MUX



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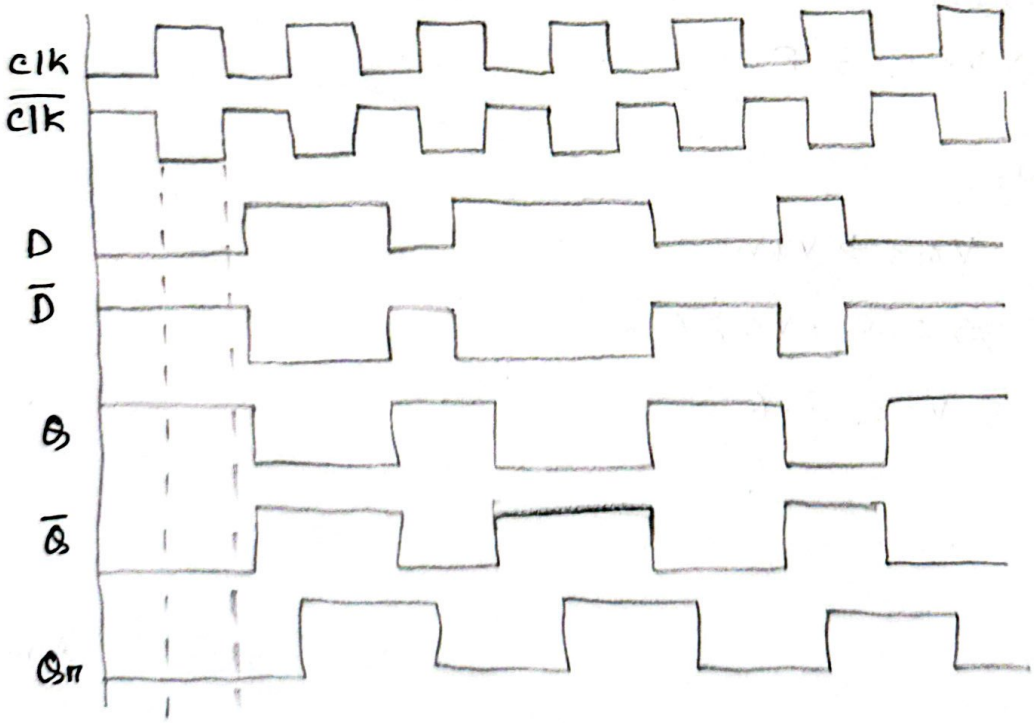
b) Clk is high (1) and $\overline{clk}$ is low (0), transmission gate ON. Clk is low (0), the transmission gate is off, 2nd trans gate itself hold the value.

So, Device 1 is a positive level triggered D-latch.

characteristic table

Enable, E	Data, D	Output, Q_{n+1}
0	X	Q_n
1	0	0
1	1	1

c) positive-edge triggered flip-flop



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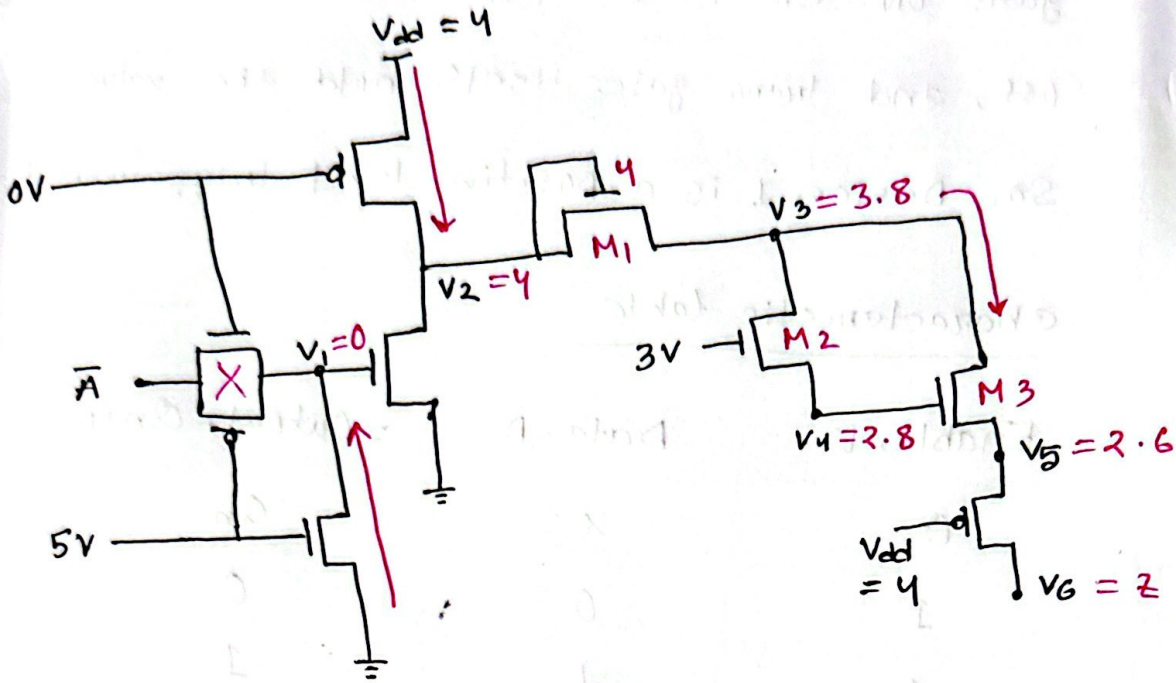
d)

$V_{tn} = |V_{tp}| = 0.2V$



M1

$V_{GS} > V_{tn}$

$\Rightarrow 4 - V_S > 0.2$

$\Rightarrow V_S \leq 3.8$

M2

$V_{GS} > V_{tn}$

$\Rightarrow 3 - V_S > 0.2$

$\Rightarrow V_S \leq 2.8$

M3

$V_{GS} > V_{tn}$

$\Rightarrow 2.8 - V_S > 0.2$

$\Rightarrow V_S \leq 2.6$

Azmani

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